GATE ME 90 Days Challenge

Day 40/90 · Practice Set with Detailed Solutions

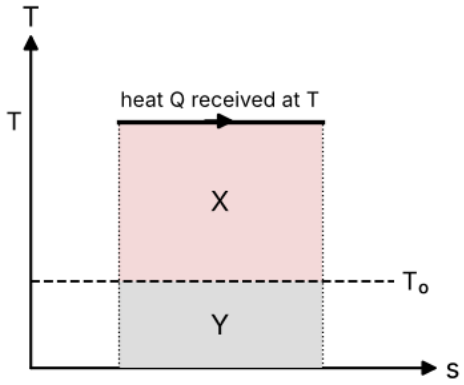
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Thermodynamics	**Availability & Relations**	**65 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on available and unavailable energy, exergy of heat, closed systems and flow streams, irreversibility (Gouy–Stodola) and second-law efficiency, the Maxwell relations, Clapeyron and Clausius–Clapeyron equations, $c_p - c_v$ relations and the Joule–Thomson coefficient. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Heat Q is received reversibly at a constant temperature T, as shown on the T–s diagram; the environment is at T_0. Which statement about the shaded areas is correct?



(A) X is the unavailable energy and Y the available energy

(B) X is the available energy and Y the unavailable energy

(C) X + Y is the available energy

(D) Y is the irreversibility of the process

Q2 NAT

2000 kJ of heat is supplied from a reservoir at 800 K; the surroundings are at 300 K. The available energy (in kJ) is _____

Q3 NAT

1000 kJ of heat flows from a reservoir at 1200 K to a reservoir at 600 K. With the surroundings at 300 K, the loss of available energy (in kJ) is _____

Q4 MCQ

The Maxwell relation obtained from the Gibbs function ($dg = v\,dp - s\,dT$) is

(A) $(\partial s/\partial p)_T = -(\partial v/\partial T)_p$

(B) $(\partial s/\partial v)_T = (\partial p/\partial T)_v$

(C) $(\partial T/\partial p)_s = (\partial v/\partial s)_p$

(D) $(\partial s/\partial p)_T = (\partial v/\partial T)_p$

Q5 NAT

3 kg of air (c_p = 1.005 kJ/kg·K) is cooled at constant pressure from 800 K to the surroundings temperature of 300 K. The available energy of the heat given out (in kJ, round off to one decimal place) is _____

Q6 MCQ

A real gas is throttled through a valve and its temperature falls. At that state its Joule–Thomson coefficient $\mu = (\partial T/\partial p)_h$ is

(A) zero

(B) negative

(C) positive, and the state lies below the inversion curve

(D) infinite

Q7 NAT

At 200 °C, saturated water has v_f = 0.001157 m³/kg, v_g = 0.12736 m³/kg and h_{fg} = 1940.7 kJ/kg. The slope dp/dT of the saturation curve at 200 °C from the Clapeyron equation (in kPa/K, round off to two decimal places) is _____

Q8 MCQ

Which statement about the specific heats of a substance is correct?

(A) $c_p - c_v = R$ for every substance

(B) $c_p < c_v$ for water between 0 °C and 4 °C

(C) $c_p = c_v$ for an ideal gas

(D) $c_p - c_v = R$ for an ideal gas and is nearly zero for liquids and solids

Q9 NAT

Air (an ideal gas, R = 0.287 kJ/kg·K) is throttled adiabatically from 800 kPa to 200 kPa. With T_0 = 300 K, the exergy destroyed per kg of air (in kJ/kg, round off to one decimal place) is _____

Q10 MCQ

A refrigerator keeps a cold space at −20 °C and rejects heat to a kitchen at 30 °C. Its actual COP is 2.5. Its second-law efficiency is about

(A) 49 %

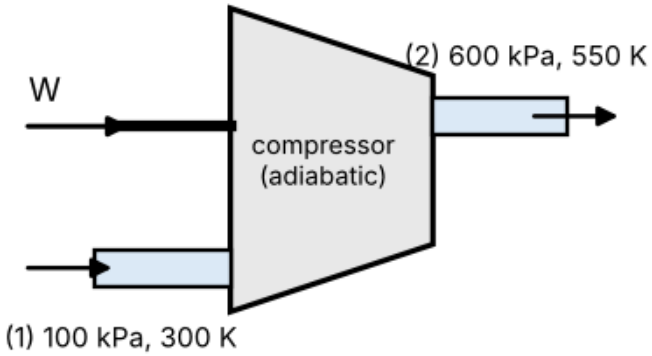
(B) 25 %

(C) 2.5 %

(D) 100 %

Q11 NAT

Air (c_p = 1.005 kJ/kg·K, R = 0.287 kJ/kg·K) is compressed in the adiabatic steady-flow compressor shown from 100 kPa, 300 K to 600 kPa, 550 K. With T_0 = 300 K, the exergy destroyed per kg of air (in kJ/kg, round off to one decimal place) is _____



Q12 MCQ

For a closed system that can exchange heat only with the atmosphere at T_0 and p_0 and ends at T_0 and p_0, the maximum useful work obtainable equals the decrease in its

(A) internal energy
(B) Gibbs function, G = H − TS
(C) Helmholtz function, A = U − TS
(D) enthalpy

Q13 NAT

Air flows steadily at 500 K and 400 kPa with negligible velocity. With the dead state at 300 K and 100 kPa (c_p = 1.005 kJ/kg·K, R = 0.287 kJ/kg·K), its specific flow exergy (in kJ/kg, round off to one decimal place) is _____

Q14 MCQ

Using the relation $(\partial u/\partial v)_T = T(\partial p/\partial T)_v - p$ for a van der Waals gas, $(p + a/v^2)(v - b) = RT$, the value of $(\partial u/\partial v)_T$ is

(A) 0
(B) RT/(v − b)
(C) a/v^2
(D) b/v^2

Q15 MCQ

Water boils at 100 °C at 101.325 kPa, with h_{fg} = 2257 kJ/kg (assume constant) and R = 0.4615 kJ/kg·K for the vapour. Using the Clausius–Clapeyron equation, the boiling point at a hill station where the pressure is 70 kPa is closest to

(A) 80 °C
(B) 90 °C
(C) 95 °C
(D) 100 °C

Answer Key

Q	Type	Answer
Q1	MCQ	(B) X is the available energy and Y the unavailable energy
Q2	NAT	1250
Q3	NAT	250
Q4	MCQ	(A) $(\partial s/\partial p)_T = -(\partial v/\partial T)_p$
Q5	NAT	620.3 (accept 616.6 to 624.0)
Q6	MCQ	(C) positive, and the state lies below the inversion curve
Q7	NAT	32.50 (accept 32.34 to 32.66)
Q8	MCQ	(D) $c_p - c_v = R$ for an ideal gas and is nearly zero for liquids and solids
Q9	NAT	119.4 (accept 118.4 to 120.4)
Q10	MCQ	(A) 49 %
Q11	NAT	28.5 (accept 28.2 to 28.8)
Q12	MCQ	(B) Gibbs function, G = H − TS
Q13	NAT	166.3 (accept 165.3 to 167.4)
Q14	MCQ	(C) a/v^2
Q15	MCQ	(B) 90 °C

Detailed Solutions

Q1 **Answer: (B) X is the available energy and Y the unavailable energy**

Concept: Of heat Q received at T, the most that can be converted to work is $Q(1 - T_0/T)$, the Carnot fraction with the environment as sink.

Area X (between T and T_0) = $(T - T_0)\Delta s$ = **available energy**

Area Y (below T_0) = $T_0\Delta s$ = **unavailable energy**, rejected to the environment even by a reversible engine.

Exam tip: Unavailable energy = $T_0 \times \Delta S$ of the heat source; it is never zero unless $T_0 = 0$ K.

Q2 **Answer: 1250**

Concept: $AE = Q(1 - T_0/T)$ for heat drawn from a reservoir at constant T.

$AE = 2000 \times (1 - 300/800)$ = **1250 kJ**; unavailable = 750 kJ

Exam tip: This is the work a Carnot engine would give using the surroundings as its sink.

Q3 **Answer: 250**

Concept: Available energy before: $Q(1 - T_0/T_1)$; after: $Q(1 - T_0/T_2)$. The difference is T_0S_{gen}.

AE at 1200 K = $1000(1 - 300/1200)$ = 750 kJ; AE at 600 K = $1000(1 - 300/600)$ = 500 kJ

Loss = 750 − 500 = **250 kJ** = $300 \times 1000(1/600 - 1/1200)$ ✓

Exam tip: Heat 'degrades' when it flows down a temperature gradient: same energy, less exergy.

Q4 **Answer: (A) $(\partial s/\partial p)_T = -(\partial v/\partial T)_p$**

Concept: For an exact differential dz = M dx + N dy, $(\partial M/\partial y)_x = (\partial N/\partial x)_y$.

With dg = v dp − s dT: $(\partial v/\partial T)_p = -(\partial s/\partial p)_T$

So $(\partial s/\partial p)_T = -(\partial v/\partial T)_p$.

Exam tip: Memory aid: the four relations come from du, dh, da, dg; the minus sign appears for du and dg.

Q5 **Answer: 620.3 (accept 616.6 to 624.0)**

Concept: For a finite source the temperature falls as heat leaves, so AE = Q − $T_0\Delta S$ (not $Q(1 - T_0/T)$).

$Q = mc_p(T_1 - T_0) = 3 \times 1.005 \times 500 = 1507.5$ kJ

$T_0\Delta S = 300 \times 3 \times 1.005 \times \ln(800/300) = 887.2$ kJ

AE = 1507.5 − 887.2 = **620.3 kJ**

Exam tip: Using the initial temperature 800 K in the Carnot factor would give 942 kJ, an overestimate.

Q6 **Answer: (C) positive, and the state lies below the inversion curve**

Concept: Throttling is isenthalpic and the pressure falls (dp < 0).

A temperature drop (dT < 0) with dp < 0 means μ = dT/dp > 0.

μ > 0 inside (below) the inversion curve: **positive**. This is how gases are liquefied in the Linde process.

Exam tip: Ideal gas: μ = 0. Hydrogen and helium at room temperature have μ < 0 (they warm on throttling) and must be pre-cooled.

Q7 **Answer: 32.50 (accept 32.34 to 32.66)**

Concept: Clapeyron: $dp/dT = h_{fg}/(T\, v_{fg})$, exact for any phase change.

$v_{fg} = 0.12736 - 0.001157 = 0.126203$ m³/kg; T = 473.15 K

$dp/dT = 1940.7/(473.15 \times 0.12620) =$ **32.50 kPa/K**

Exam tip: kJ/m³ = kPa, so with h in kJ/kg and v in m³/kg the slope comes out directly in kPa/K.

Q8 **Answer: (D) $c_p - c_v = R$ for an ideal gas and is nearly zero for liquids and solids**

Concept: General relation: $c_p - c_v = Tv\beta^2/\kappa$.

Ideal gas: β = 1/T, κ = 1/p → $c_p - c_v = pv/T = R$.

Liquids and solids have very small β, so $c_p \approx c_v$, which is why a single 'c' is quoted for them.

So the correct choice is **(D)**.

Exam tip: Since κ > 0 always, c_p can never be less than c_v.

Q9 **Answer: 119.4 (accept 118.4 to 120.4)**

Concept: Throttling an ideal gas: h and T stay constant, but entropy rises.

$s_{gen} = R \ln(p_1/p_2) = 0.287 \ln 4 = 0.39787$ kJ/kg·K

$i = T_0 s_{gen} = 300 \times 0.39787 =$ **119.4 kJ/kg**

Exam tip: No energy is lost in a valve, but exergy is — that is why expansion valves are replaced by turbines in large plants.

Q10 **Answer: (A) 49 %**

Concept: $\eta_{II} = COP/COP_{rev}$.

$COP_{rev} = T_L/(T_H - T_L) = 253.15/50 = 5.06$

$\eta_{II} = 2.5/5.06 =$ **0.494** ≈ 49 %

Exam tip: Second-law efficiency compares with the best possible device between the same temperatures; it can never exceed 1.

Q11 **Answer: 28.5 (accept 28.2 to 28.8)**

Concept: Adiabatic device: $s_{gen} = s_2 - s_1$; Gouy–Stodola: $i = T_0 s_{gen}$.

$s_2 - s_1 = 1.005 \ln(550/300) - 0.287 \ln 6 = 0.60915 - 0.51424 = 0.09493$ kJ/kg·K

$i = 300 \times 0.09493 =$ **28.5 kJ/kg**

Actual work = 1.005 × 250 = 251.2 kJ/kg, so about 11 % of the input is destroyed.

Exam tip: The minimum (reversible) work = actual work − $T_0 s_{gen}$ for an adiabatic compressor.

Q12 **Answer: (B) Gibbs function, G = H − TS**

Concept: Useful work excludes the work done against the atmosphere, $p_0 \Delta V$.

At constant T_0 and p_0, $W_{useful,max} = -\Delta G = G_1 - G_2$.

At constant T and V (no atmospheric work), the maximum work is $-\Delta A$ (Helmholtz function A = U − TS).

So the answer is the **Gibbs function**.

Exam tip: Helmholtz → constant T and V; Gibbs → constant T and p.

Q13 **Answer: 166.3 (accept 165.3 to 167.4)**

Concept: Flow exergy $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$.

$h - h_0 = 1.005 \times 200 = 201.0$ kJ/kg

$s - s_0 = 1.005 \ln(5/3) - 0.287 \ln 4 = 0.51338 - 0.39787 = 0.11551$ kJ/kg·K

$\psi = 201.0 - 300 \times 0.11551 =$ **166.3 kJ/kg**

Exam tip: Flow exergy uses h (it includes flow work); non-flow exergy uses u + $p_0 v$.

Q14 **Answer: (C) a/v²**

Concept: From the van der Waals equation, $p = RT/(v - b) - a/v^2$, so $(\partial p/\partial T)_V = R/(v - b)$.

$T(\partial p/\partial T)_V - p = RT/(v - b) - [RT/(v - b) - a/v^2] =$ **a/v^2**

For an ideal gas (a = 0) it is zero: u depends on T only (Joule's law).

Exam tip: Internal energy of a real gas rises with volume at fixed T because molecules move apart against attraction.

Q15 **Answer: (B) 90 °C**

Concept: Clausius–Clapeyron (ideal vapour, v_f neglected): $\ln(p_2/p_1) = (h_{fg}/R)(1/T_1 - 1/T_2)$.

$1/T_2 = 1/373.15 - (0.4615/2257)\ln(70/101.325) = 0.0026799 + 0.0000756$

$T_2 = 362.9$ K = **89.8 °C ≈ 90 °C**

Exam tip: Lower pressure → lower boiling point; this is why a pressure cooker (higher p) cooks faster.

What next?

Day 40 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 41/90: Heat Transfer, 1-D Conduction. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.